Module X
UNET

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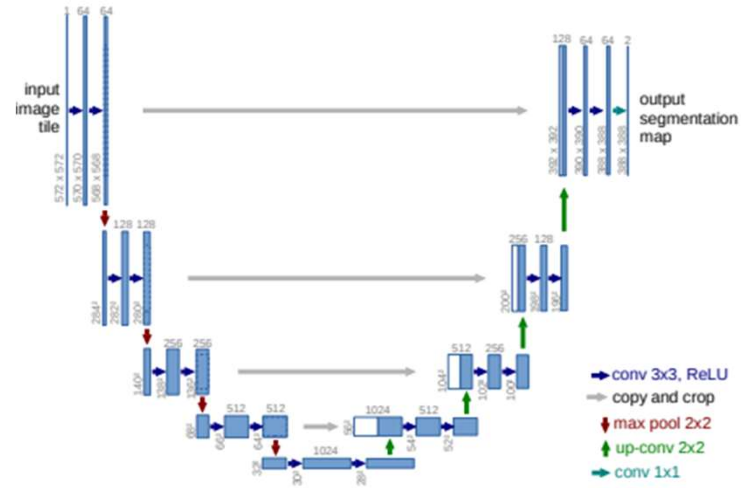
UNET

- U-Shaped Architecture:
- The U-Net architecture is symmetric and comprises two parts:
  - a contracting path (encoder)
  - expansive path (decoder).
- This U-shape gives the network its name.

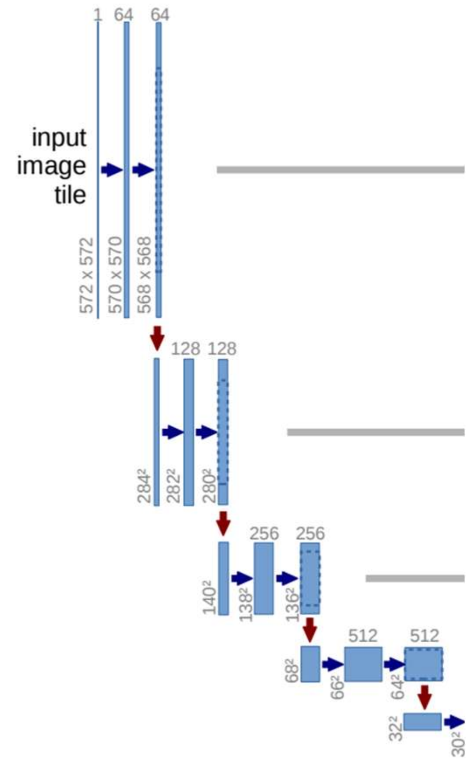


Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation.

The contracting path, Encoder

Encoder:

- It consists of repeated applications of convolution and max-pooling operations.
- Each step in the encoder halves the spatial dimensions while doubling the number of feature channels.

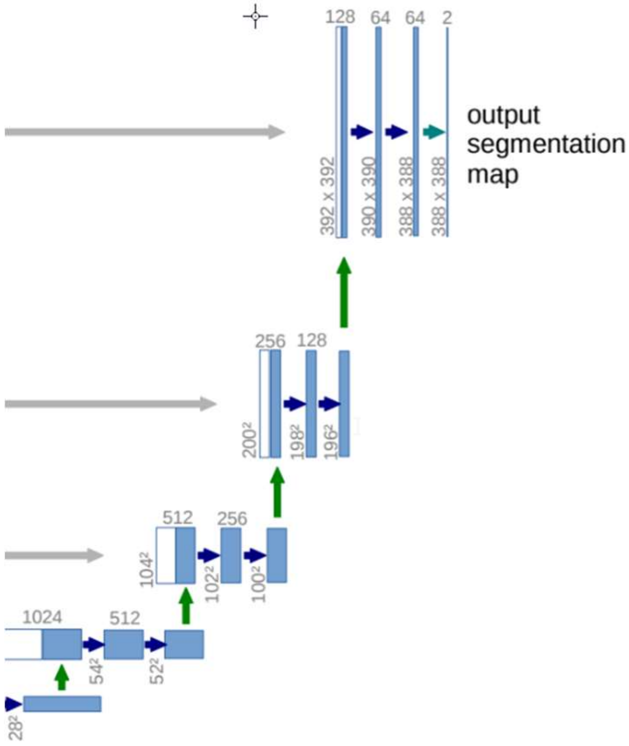


The expansive path, Decoder

Decoder:

- The expansive path consists of upsampling operations and concatenations with corresponding high-resolution features from the contracting path.

- This helps to recover spatial information lost during downsampling.

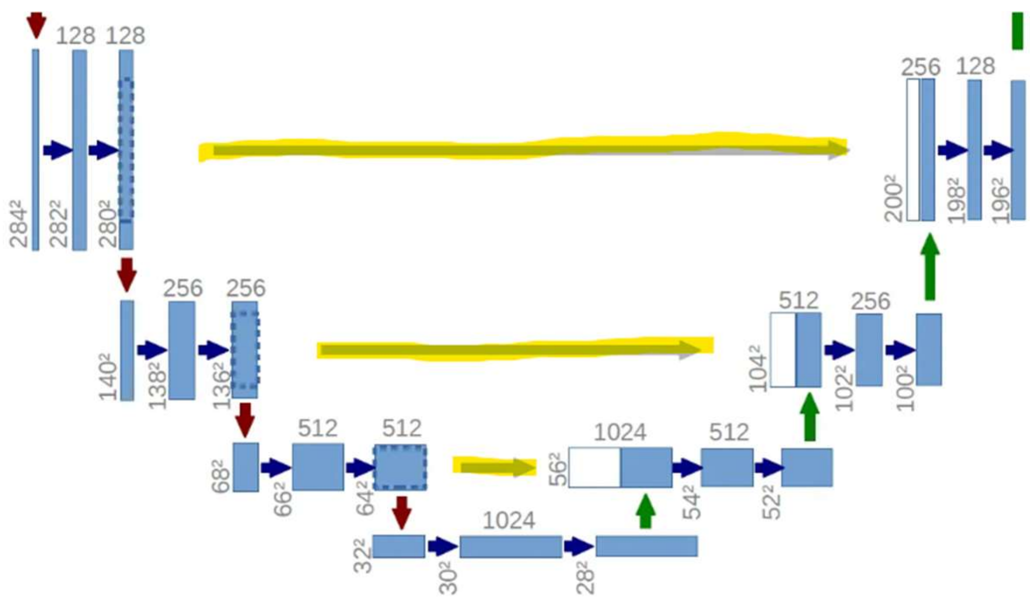


Bottleneck

- At the bottom of the U, there's a bottleneck layer that captures the most compact representation of the input.

Skip Connections

- The U-Net uses skip connections that concatenate the feature maps from the encoder to the corresponding feature maps in the decoder. This helps in better localization and reduces information loss.



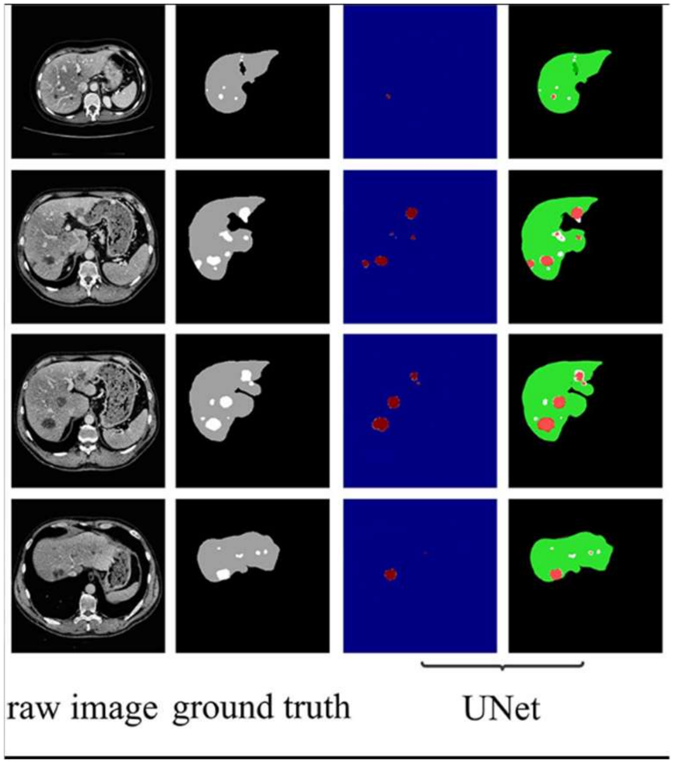
Why Skip Connections Help:

- **Preserve Information:** They help retain important spatial information from the input image that might otherwise be lost through successive downsampling layers.
- **Improve Localization**: By providing high-resolution feature maps directly to the decoder, they enhance the network's ability to accurately localize and segment fine details in the input image.
- **Gradient Flow**: They facilitate better gradient flow during backpropagation, making it easier to train deep networks by alleviating the vanishing gradient problem.

Why U-Net Enables Segmentation

- **Segmentation**: Assigns labels to each pixel, dividing images into meaningful regions.

- U-Net excels at this due to its ability to capture both global context and local details through an encoder-decoder architecture with skip connections.



How U-Net Captures Context & Detail

- **Encoder** (Downsampling) → Captures Global Features
  - Uses convolutions & pooling to extract high-level features like shapes & textures.
  - Reduces spatial resolution but increases feature depth for abstract understanding.
- **Decoder** (Upsampling) → Restores Spatial Structure
  - Expands feature maps to recover image resolution.
  - Without additional detail, segmentation may become blurry.
- **Skip Connections** → Preserve Fine Details
  - Directly transfer high-resolution features from encoder to decoder.
  - Prevents information loss & ensures precise boundary localization.

Key Components for Segmentation

- **Encoder** (What is in the image?)
  - Learns to differentiate textures & structures through hierarchical feature extraction.
- **Bottleneck** (Compressed Representation)
  - Captures essential patterns in a latent space for robust decision-making.
- **Decoder** (Where are the objects?)
  - Up samples feature maps to map the segmentation back to the original image size.
- **Skip Connections** (Bringing Back Details)
  - Directly pass fine-grained spatial info to decoder, preventing loss of small structures.